

# **BOARDLINK**

## **Introduction & Project Name**

The individual project titled “BoardLink – Color-Coded Task Indicator” is an enhancement of the existing BoardLink system — a project and task management platform that helps users plan, track, and organize their work efficiently.

While BoardLink allowed effective creation and management of tasks, it lacked a visual alert mechanism that could help users quickly identify tasks based on their urgency or due date status.

To address this limitation, the Color-Coded Task Indicator feature was developed.

This feature visually represents task deadlines using color changes — allowing users to instantly recognize overdue, upcoming, and safely scheduled tasks.

By introducing this intuitive color-based system, BoardLink enhances project tracking, time management, and overall productivity.

## **Identifying Business Problem from the Group Project**

In the original BoardLink project, users could manage boards and tasks effectively. However, there was no quick visual way to understand which tasks were urgent, nearing deadlines, or overdue.

This created several real-world challenges:

- Users had to manually check due dates for each task, wasting valuable time.
- Overdue tasks often went unnoticed, affecting project timelines.
- Lack of visual cues made it difficult to prioritize tasks efficiently.
- Managers couldn’t identify task urgency at a glance.

The new Color-Coded Task Indicator feature solves these issues by providing an immediate, color-based representation of task deadlines — improving clarity, focus, and decision-making.

## **Objective of the New Feature**

The BoardLink – Color-Coded Task Indicator feature was designed with the following key objectives:

1. Enhance Visual Awareness: Provide color indicators (Red, Yellow, Green) based on task due dates.
2. Improve Time Management: Allow users to identify and prioritize urgent tasks quickly.

3. Automate Deadline Alerts: Automatically update task colors without manual checks.
4. Seamless Integration: Embed the feature smoothly within the existing BoardLink dashboard.
5. Boost Productivity: Enable faster decision-making and better focus on critical tasks.

## **Requirements**

### **Functional Requirements**

- Automatically change task card colors based on due date:
  - Red: Task overdue (past due date).
  - Yellow: Task due within 2 days.
  - Green: Task due in more than 2 days.
- Update task color dynamically on page load or refresh.
- Store and retrieve due dates securely from the PostgreSQL database.
- Reflect real-time changes when a user edits a task's due date.
- Display visual indicators on all project boards for clear task visibility.

### **Non-Functional Requirements**

- A responsive and user-friendly UI integrated with the BoardLink dashboard.
- Low-latency rendering to ensure color updates appear instantly.
- Lightweight JavaScript logic for fast color determination.
- Secure data integration with Flask and PostgreSQL backend.
- Scalable for future enhancements such as email or popup deadline reminders.

### **Tools & Technologies**

- Frontend: HTML5, CSS3, JavaScript
- Backend: Flask (Python)
- Database: PostgreSQL
- Frameworks: SQLAlchemy ORM, Jinja2 Templates

## Technical Work and Analysis

The Color-Coded Task Indicator feature was implemented by enhancing the Task model in BoardLink to include due date tracking and real-time visual rendering through Flask templates and JavaScript.

Whenever the BoardLink dashboard is loaded:

- The system retrieves all tasks and their due dates from the PostgreSQL database.
- JavaScript compares each task's due date with the current date.
- Based on the time difference, the background color of the task card changes automatically:
  - Red if the task is overdue.
  - Yellow if the due date is within the next 2 days.
  - Green if the due date is more than 2 days away.

Additionally:

- Flask ensures accurate time and date synchronization for all users.
- SQLAlchemy ORM securely handles all task-related data transactions.
- The design blends perfectly into the existing task board layout without affecting other features.
- The system is fully dynamic — any due date changes trigger immediate color updates upon reload.

## Solving the Business Problem

The BoardLink – Color-Coded Task Indicator feature effectively addresses the earlier problem of missed or delayed task identification.

It introduces a smart, visual management layer that keeps users aware of approaching deadlines and overdue work.

With this feature:

- Users can instantly recognize which tasks need immediate attention.
- Managers gain a quick overview of project health using visual cues.
- The system promotes better organization and timely completion of tasks.
- Productivity and accountability are enhanced across all project teams.

## Result of the New Feature Update

After integrating the Color-Coded Task Indicator feature into BoardLink:

- Task cards now visually display urgency levels using colors.
- Overdue and upcoming tasks are instantly noticeable.
- Time management and prioritization have significantly improved.

- The dashboard feels more dynamic and user-friendly.
- Overall project coordination and task completion rates have increased.

## **Conclusion**

The BoardLink – Color-Coded Task Indicator feature successfully enhances the existing BoardLink system by adding a powerful visual tool for task monitoring.

It simplifies deadline tracking and ensures that users remain constantly aware of task priorities. By combining Flask, PostgreSQL, and JavaScript, this project demonstrates effective full-stack integration and intelligent front-end automation.

Overall, the Color-Coded Task Indicator transforms BoardLink into a smarter, more efficient project management system — allowing teams to stay organized, meet deadlines, and maintain productivity through clear visual insights.